10-sep-26 Progress Report

Implementation of PAUT on FPGA

Phase 1:

- ~~Estimate Element positions~~
- ~~Steering angle generator~~
- ~~Delay law calculation~~
- Fractional delay sampling and interpolation
- DAS
- Envelope detector
- Normalisation

Phase 2:

- Integration and Optimization
- Comparing the S scan with the Python S scan (we can change the coordinates from angle vs depth to x-z coordinates)

Phase 3:

- We will mostly use existing Ethernet RGMII, BRAM, and DDR3 codes
- final implementation of FPGA and comparing the output software output and quantitate metrics.

Work Division:

- Nagendra: Developing the Verilog codes and performing functional verification and implementation through simulation.
- Srinivas(B.Tech student): He will develop the top-level modules for the PAUT FPGA implementation and verify their outputs on the FPGA by displaying binary values on LEDs.
- This hardware-level verification will help us identify actual clocking issues, timing constraints, latency, and resource utilization, enabling us to further optimize the designs.
- He has already started the FPGA implementation of the EEP module.

Documentation Link:

Code Link: